



IS TURBULENT CONVECTION THE ONLY EXCITING MECHANISM OF ACOUSTIC MODES IN SOLAR-LIKE OSCILLATORS?

Eva Panetier¹, Rafael A. García², Sylvain N. Breton³,
Antonio Jiménez^{4,5}, Thierry Foglizzo²
eva.panetier@cea.fr

1. INTRODUCTION

Magnetic cycle of solar-like stars have an impact on modes properties such as **frequency, amplitude, excitation and damping**.
[e.g. Broomhall et al. (2014) and references therein]

While modes are thought to be **stochastically excited by turbulent convection**, observations reveal a power excess not fully explained by this mechanism.
[Chaplin et al. 1997, Chang & Gough 1998]

The **role of magnetic cycles in mode excitation** remains unclear — could magnetic activity features (flares, CMES,...) be responsible for this observed power excess?

CMEs

? larger characteristic size
✓ strength
? characteristic time

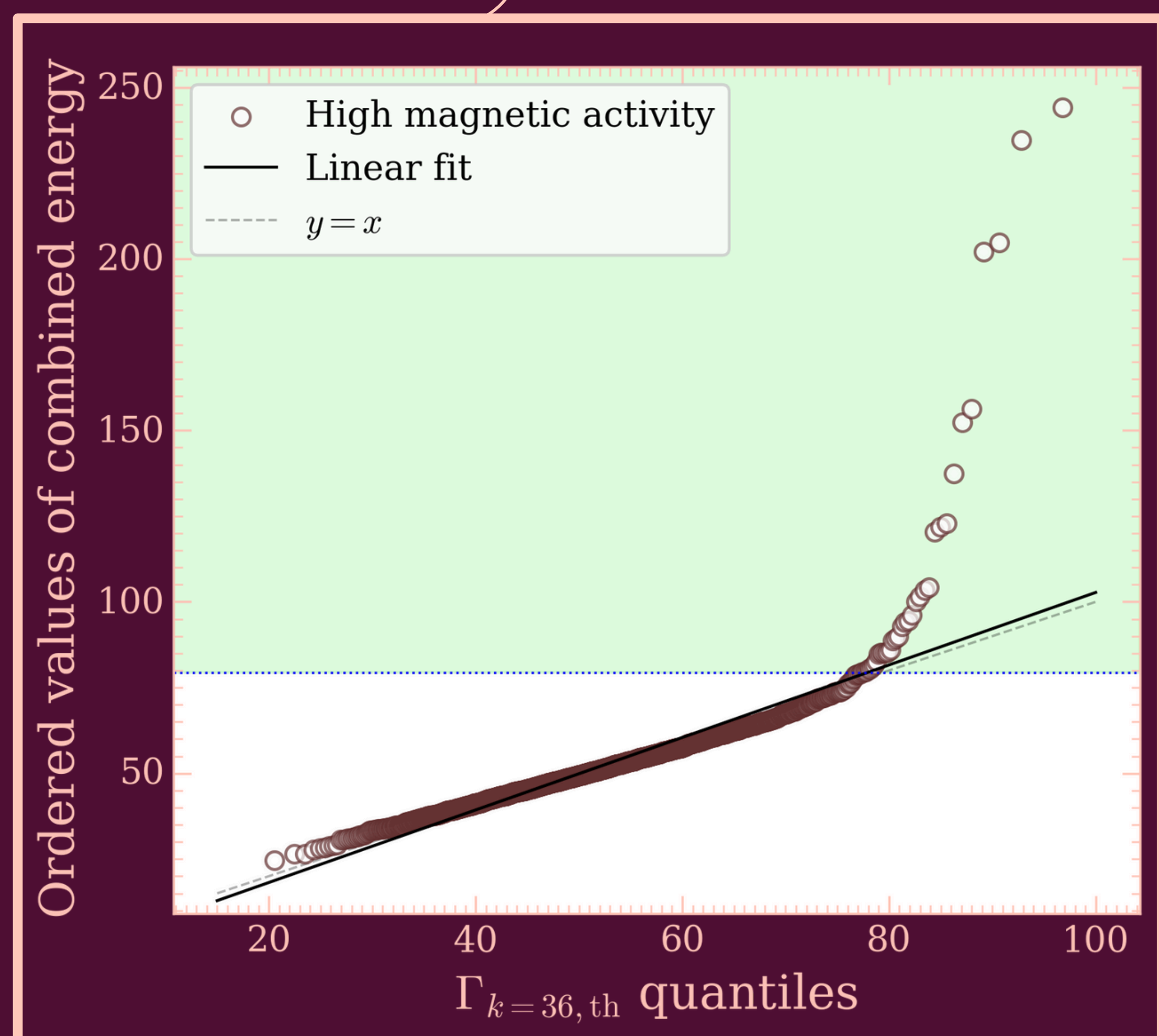
Flares

✓ small characteristic size
✗ strength
? characteristic time
[Foglizzo 1998]

4. DISTRIBUTION OF THE MODES COMBINED ENERGY

Not compatible with a stochastic excitation by turbulent convection

Theoretical probability of observing these events over all data points:
 $P < 10^{-3} \neq 1.6\%$ observed
→ **Very significant**

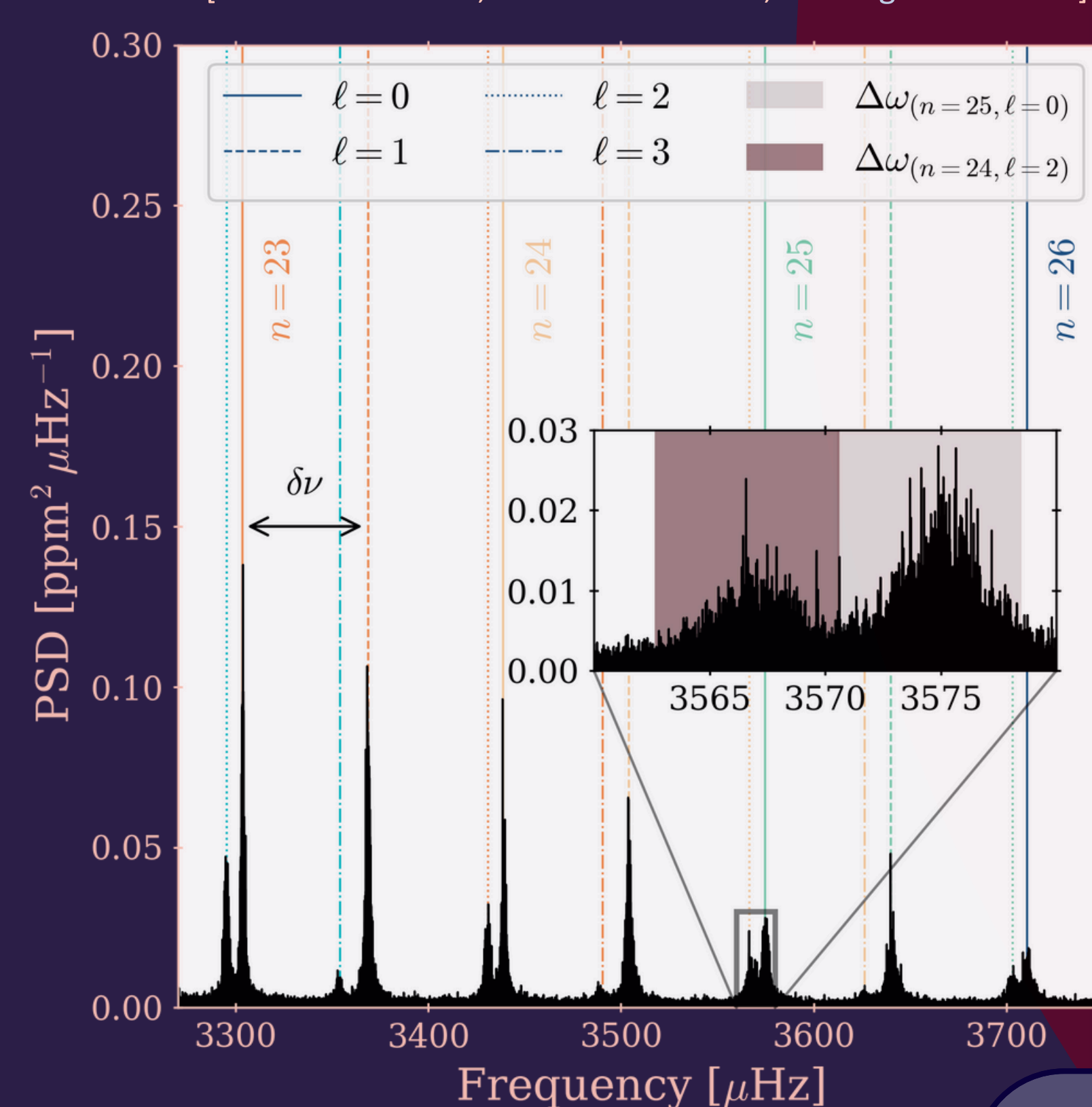


→ 108 high energy points over 6612 data points

Compatible with a stochastic excitation by turbulent convection

2. TEMPORAL EVOLUTION OF MODE ENERGY

VIRGO/SPMs and GOLF data from SoHO
[Fröhlich et al. 1995, Gabriel et al. 1995, Domingo et al. 1995]



Selection of a mode

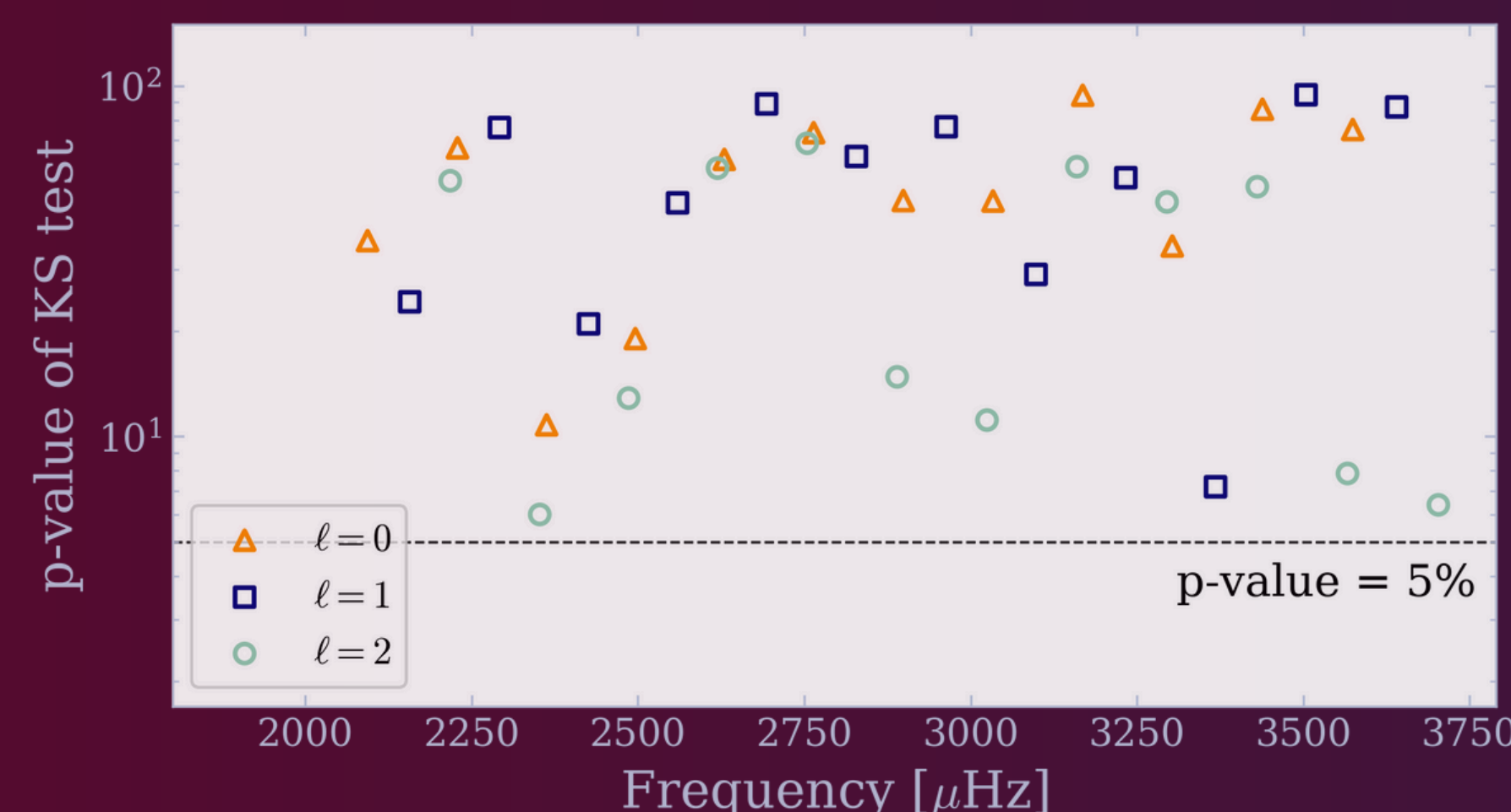
Inverse Fourier-Transform

$$E_n = 2 |f_v(t)|^2$$

Combinaison of the $\ell < 3$ and $n = 14 \rightarrow 25$ modes

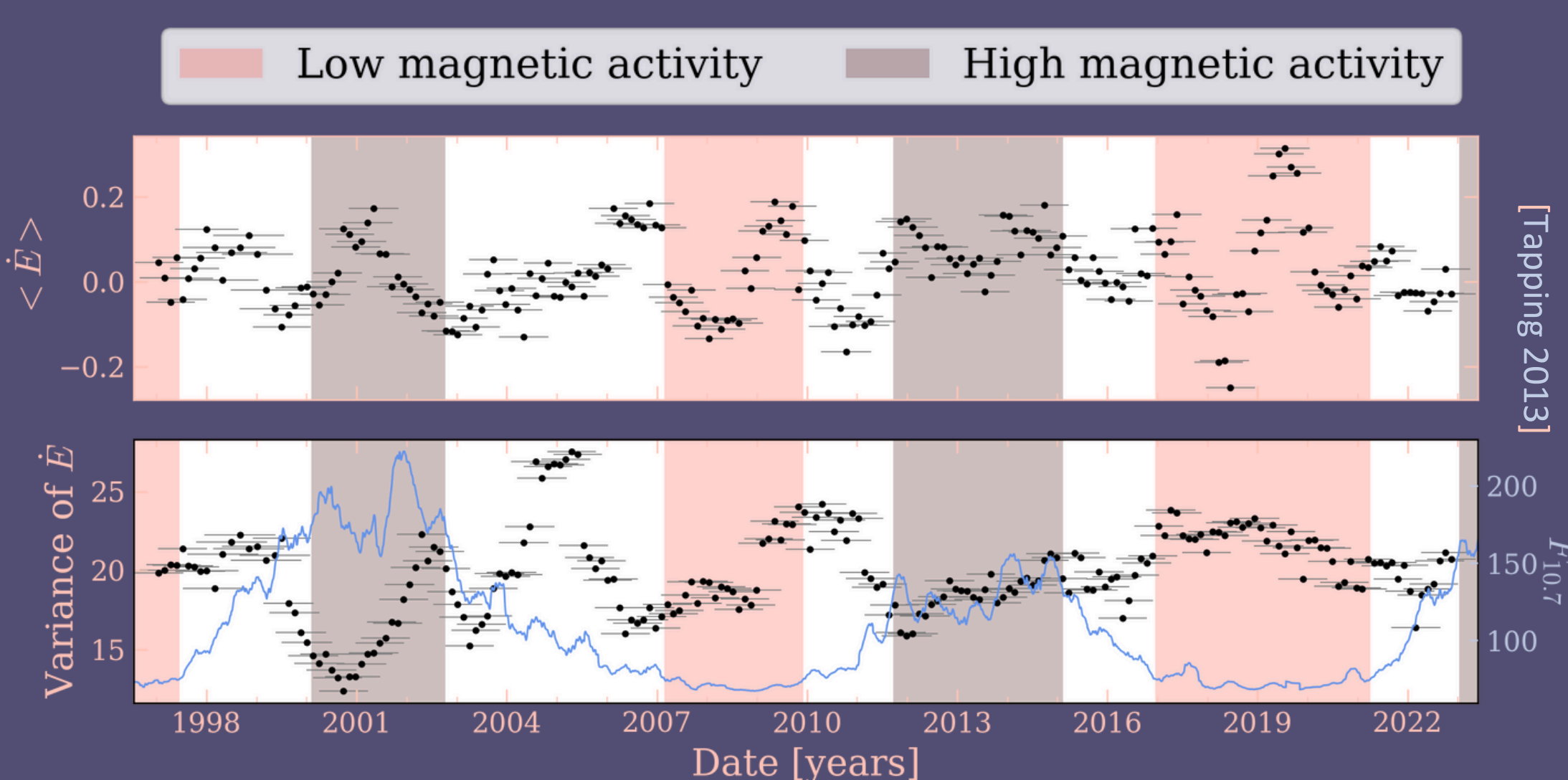
3. KOLMOGOROV-SMIRNOV STATISTICAL TESTS

Comparison of the energy distribution for each mode with theoretical exponential distributions [Nigam & Kosovichev 1998]



General behaviour of the mode energy compatible with a stochastic excitation

5. Variations in Energy supply rate



A proxy of the energy supply rate was computed: its mean and variance changes over time

¹Université Paris Cité, Université Paris-Saclay, CEA, CNRS, AIM
²Université Paris-Saclay, Université Paris Cité, CEA, CNRS, AIM
³INAF – Osservatorio Astrofisico di Catania, Italy
⁴Instituto de Astrofísica de Canarias (IAC), Tenerife, Spain
⁵Universidad de La Laguna (ULL), Departamento de Astrofísica, Tenerife, Spain

Acknowledgements. The authors want to acknowledge Leila Bessila, Adam Finley, Kiran Jain, Barbara Perri and Antoine Strugarek for useful comments and discussions. The GOLF and VIRGO/SPMs instruments on board SoHO, are a cooperative effort of many individuals to whom we are indebted. SoHO is a project of international collaboration between ESA and NASA. The authors strongly acknowledge the French space agency, CNES, for its support to GOLF since the launch of SoHO. E.P. and R.A.G. acknowledge the support from the GOLF/SoHO and PLATO Centre National D'Études Spatiales grants. S.N.B acknowledges support from PLATO ASI-INAF agreement no. 2022-28-HH.0 "PLATO Fase D".

High peaks = excitation of several modes at the same time

1 The energy supply rate changes over time [Kiefer & Broomhall 2021]

→ seems to follow a QBO-like pattern → The variance of seems to be anti-correlated with the cycle

2 There are non-stochastic characteristics in the mode excitation mechanisms

→ Observed with several instruments → Can not only be attributed to noise
→ Not automatically correlated with flares/CMEs → Magnetic cycle dependent only for the 2620-3160 Hz frequency band

A study to continue on other stars with PLATO!

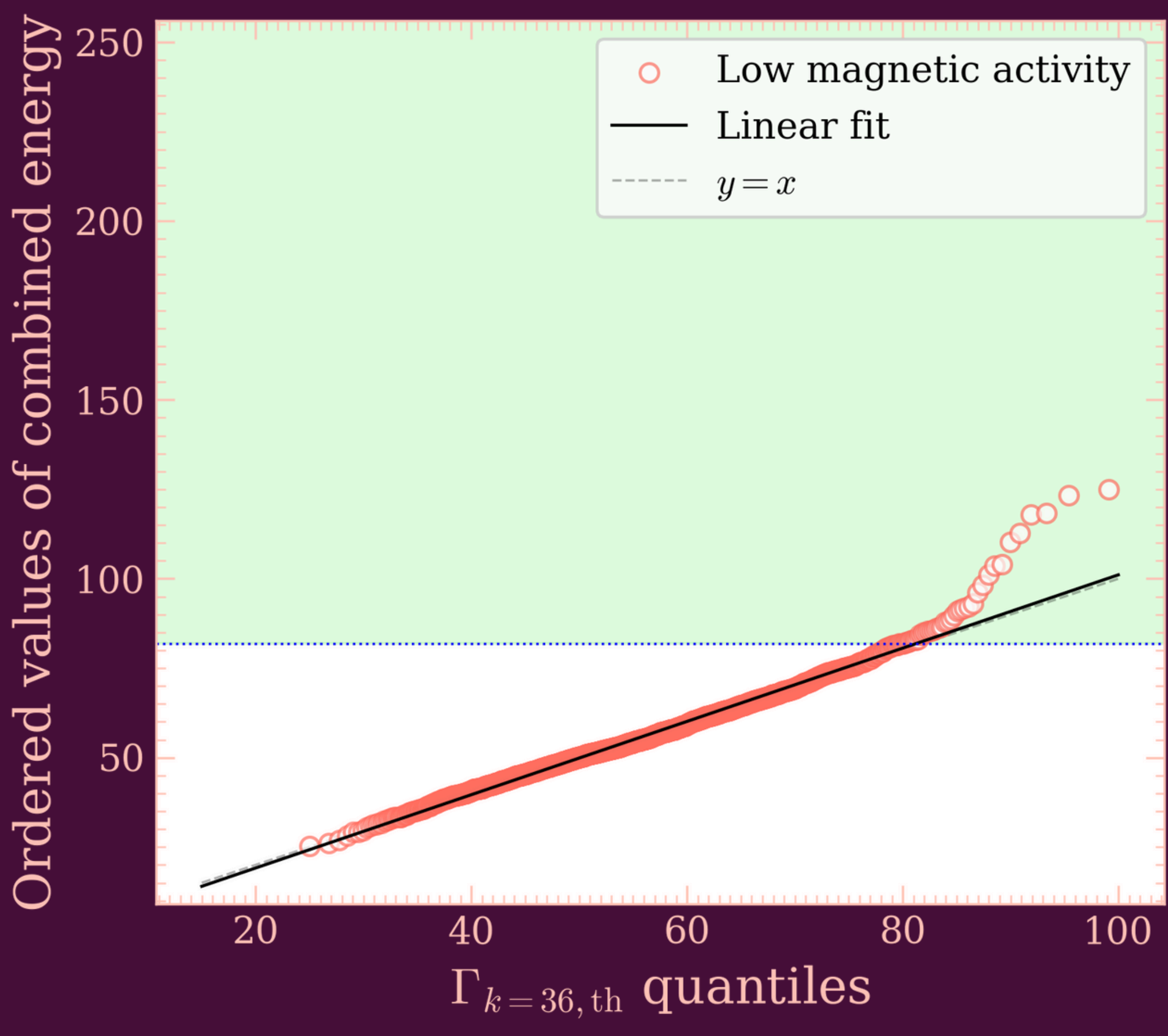
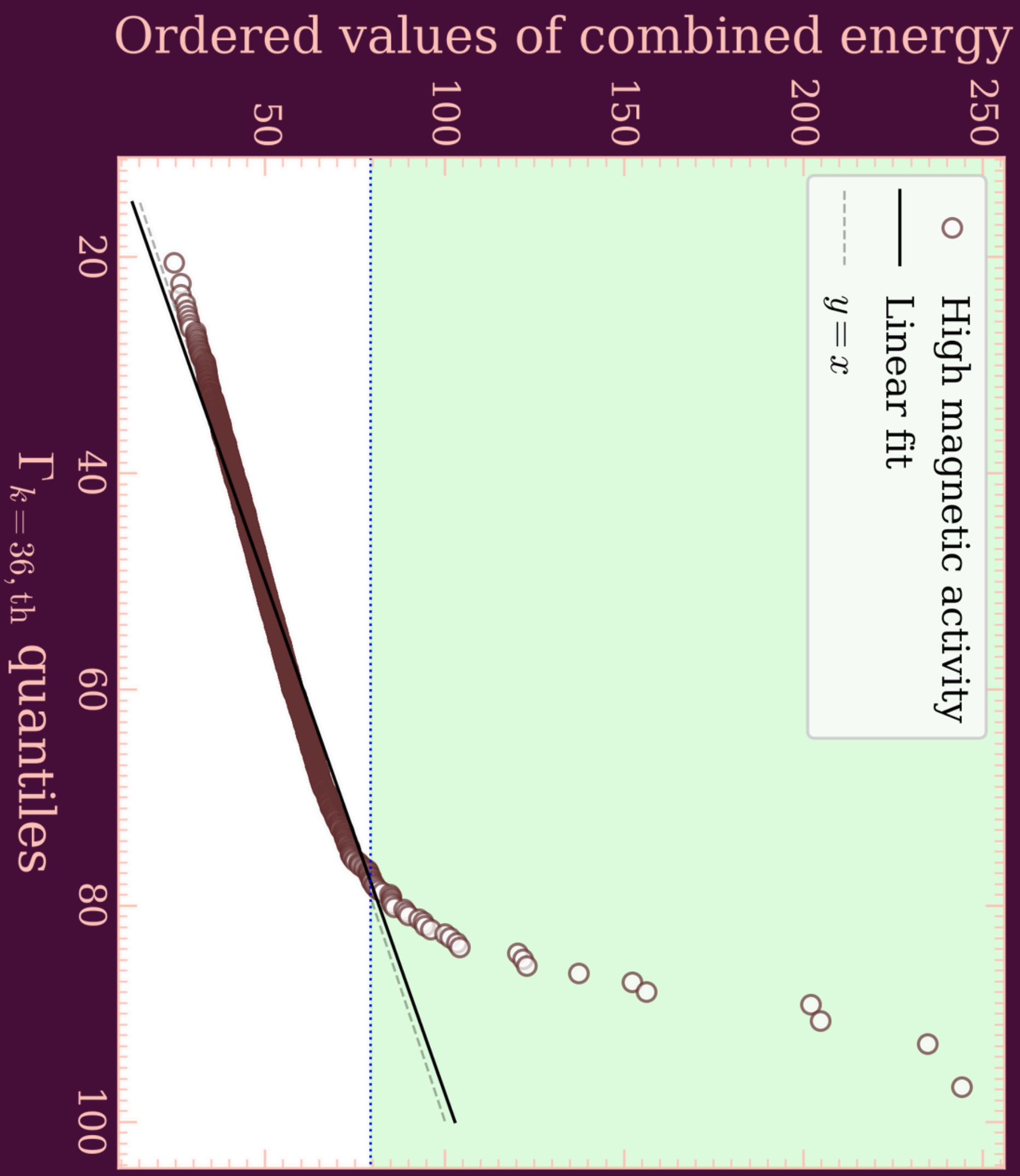
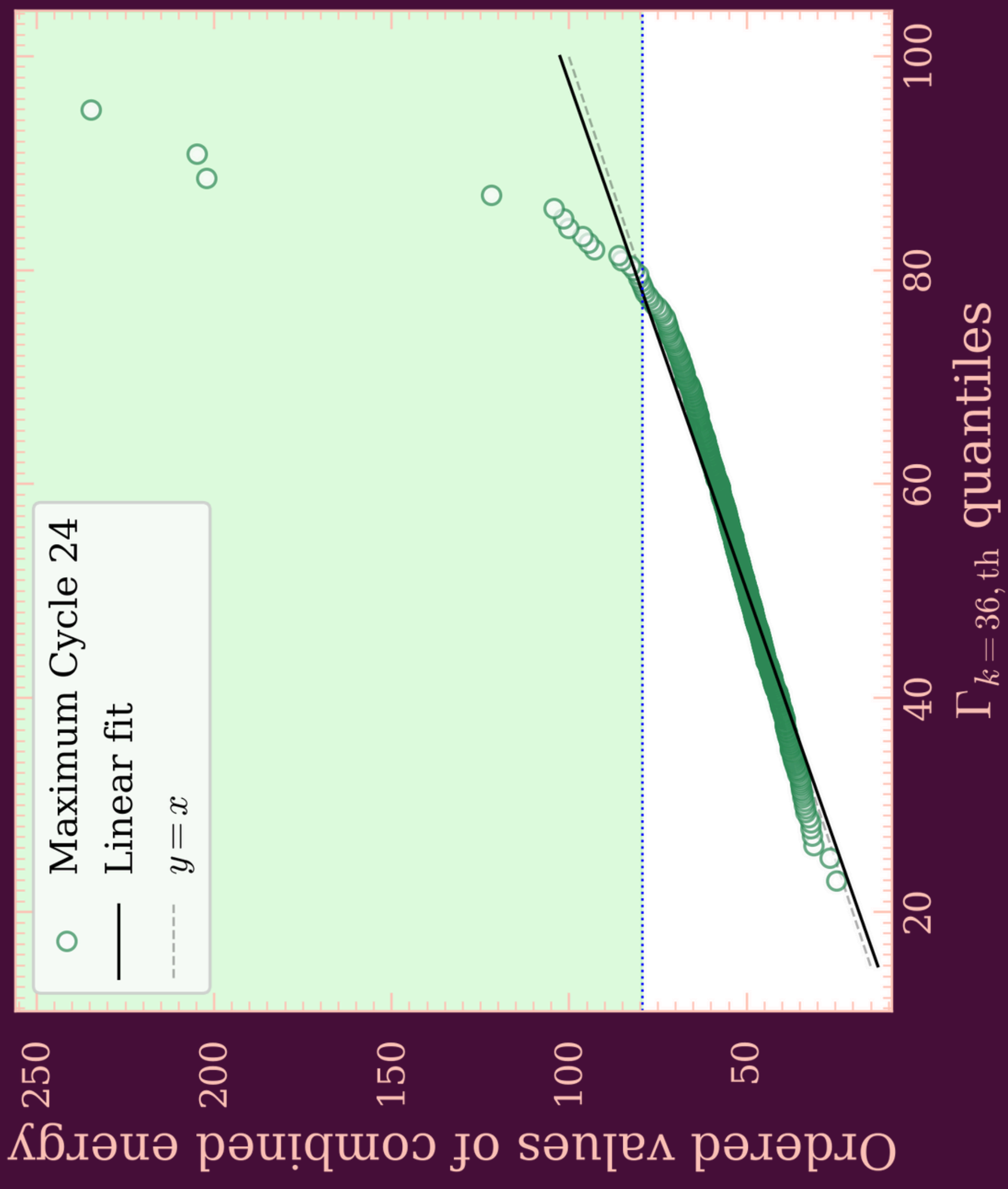
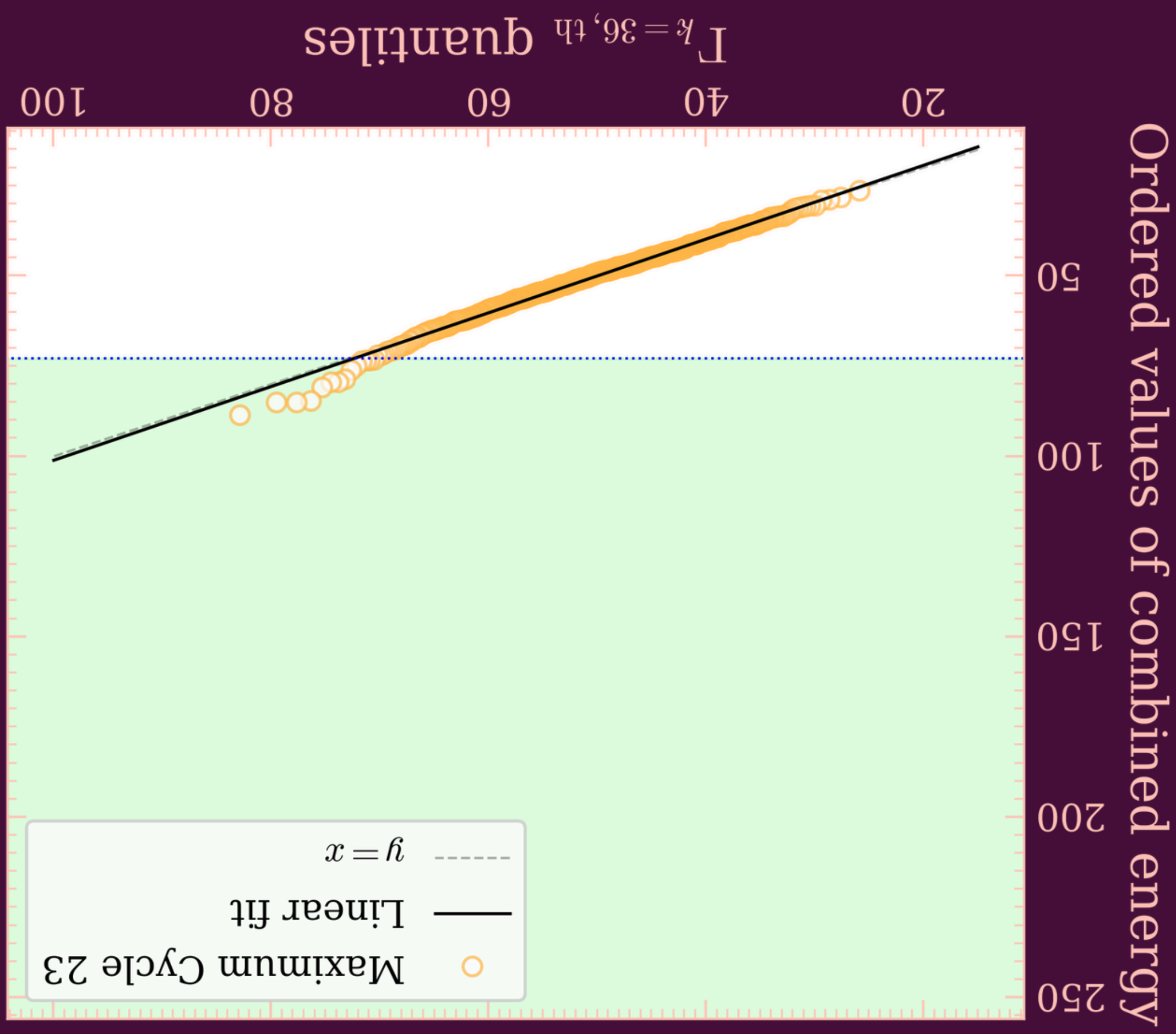


TAKE AWAY
MESSAGES

Check the paper here!



Turn the wheel to see the other Q-Q plots



Turn the wheel to see the other Q-Q plots

Turn the wheel to see the other Q-Q plots

Turn the wheel to see the other Q-Q plots